

Comparing metrics across different fine-tuning techniques for MedMCQA

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Motivation

- Full fine-tuning of a 14B model is memory-intensive — optimizer state pushes it to ~225 GB, well beyond a single GPU's memory, forcing a multi-GPU sharded setup.
- LoRA / QLoRA collapse the job back onto **one GPU** and promise near-equal quality at a **fraction of the cost** — but is that true on a medical benchmark?

How does each method's **resource cost** compare to the **accuracy** it buys?

Measured across three axes — **peak VRAM** · **GPU-hours** · **accuracy** — for full FT vs LoRA vs QLoRA, against a zero-shot baseline.

Setup: model, data, task

- **Model:** Qwen2.5-14B (instruct base)
- **Data:** MedMCQA — 4-option medical entrance-exam. We use `exp` as chain-of-thought targets
- **Training:** completion-only loss masking
- **Eval:** 1000-example held-out subset, exact answer-letter match

Fine-tuning methods

	Trainable	Base precision	GPUs
Full FT	14.8B	bf16 + ZeRO-3	4× A100-80GB
LoRA	25.2M (r16/ α 32)	bf16 frozen	1× A100-80GB
QLoRA	25.2M	4-bit NF4	1× A100-80GB

How full FT was fit?

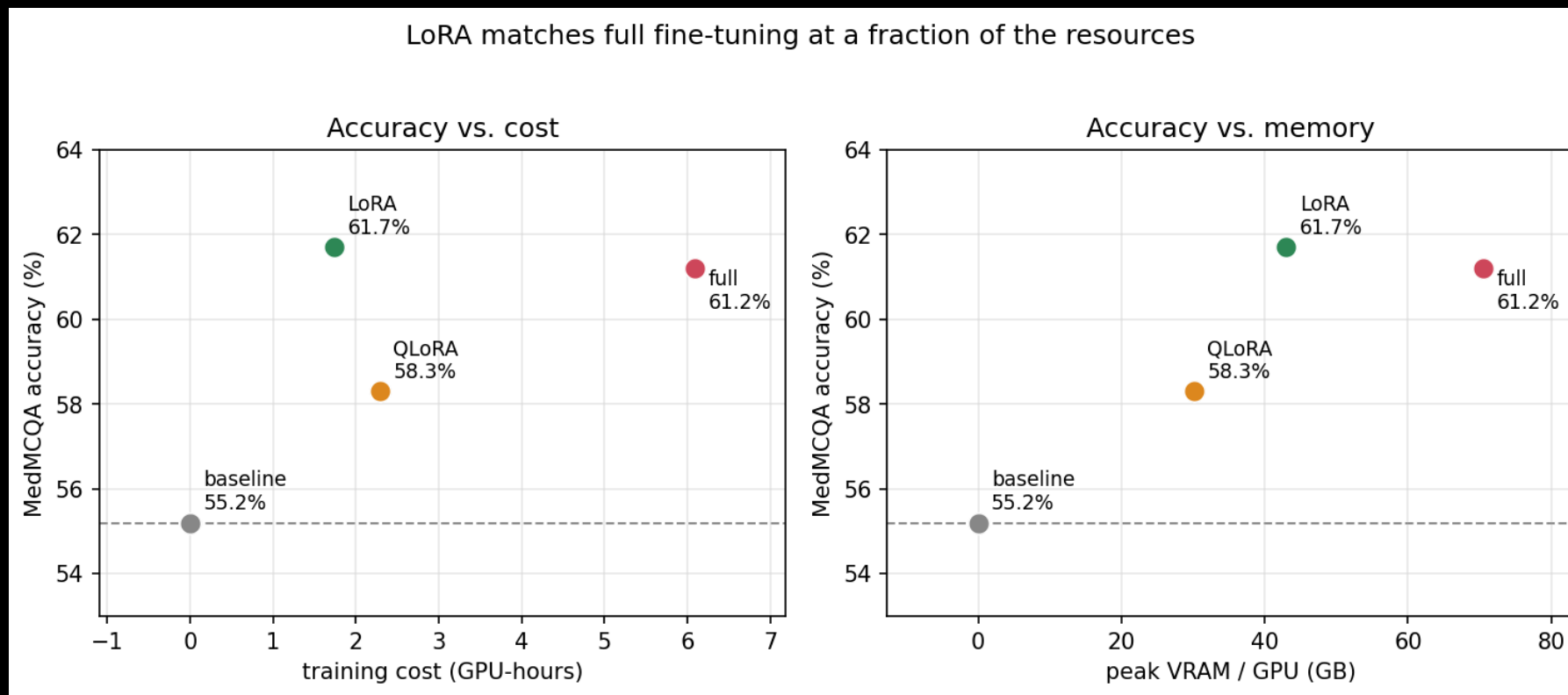
- Full-FT footprint $\approx$ **16 bytes/param** (fp32 master weights + 2 Adam moments) $\approx$ **225 GB**, exceeds most GPU's memory.
- DeepSpeed **ZeRO-3** partitions optimizer state, gradients, and parameters across 4 GPUs, leading to **70.5–82.6 GB/GPU**, just fits.

Results: accuracy

Method	Accuracy	Δ vs baseline
Baseline (no FT)	55.2%	—
Full FT	61.2%	+6.0
LoRA	61.7%	+6.5
QLoRA	58.3%	+3.1

- CoT fine-tuning adds considerable improvements on accuracy
- All methods **beat the 55.2% baseline**
- QLoRA trails LoRA by **3.4 pts**
- LoRA **ties** full FT (within $\pm 3\%$ eval noise)

Resource usage against cost



- Lora matches accuracy at the fraction of a cost.

Resource trade-off

	GPUs	Peak VRAM	tok/s	GPU-h
Full	4	70.5–82.6 GB	1125	6.09
LoRA	1	43.0–63.5 GB	981	1.74
QLoRA	1	30.1–51.8 GB	747	2.29

- QLoRA is cheapest on memory (fits a 24 GB card) but slowest (4-bit dequant overhead) and least accurate

Takeaways

- **LoRA matched full FT at ~1/3.5 the cost.**
- QLoRA trades accuracy for the smallest footprint, good fit when memory-bound.

Future work

- Equal accuracy at 10k examples could mean a shared ceiling or that full's 600× capacity was never exercised
- **Sweep LoRA rank** ($r = 8 / 16 / 32 / 64$) to find where adapter capacity starts limiting accuracy